

Northeast Sales & Inventory Performance Analysis

Vivo Smartphones | 30-Day Period (May 2026)

MIS cum Data Analyst Assessment Project | Northeast India Region

1. Key Summary

Metric	Value
Total Sell Out	24,602 units
Total Target	31,916 units
Overall Achievement %	77.08%
Branches Covered	7 (Guwahati, Silchar, Agartala, Dimapur, Imphal, Shillong, Jorhat)
Models Tracked	5 (Y11, Y21, X300FE, V70FE, Ultra)
Total Records	1,050
Data Quality Flags	9 records flagged with negative closing stock

The overall achievement stands at 77.08%, meaning roughly 23% of the monthly target was not met. No branch individually crossed 80%, but the performance gap between the best and worst branch is only about 3 percentage points — so this is a region-wide challenge rather than one specific branch pulling numbers down. Budget and entry-level models are clearly doing the heavy lifting, while the premium segment is noticeably underperforming.

2. Sales Performance Insights

The region sold 24,602 units against a target of 31,916, landing at 77.08% achievement for the 30-day period. Breaking it down by price segment gives a clearer picture of where demand is actually coming from:

Segment	Sell Out	Target	Achievement %	Contribution %
25K–35K	8,809	11,509	76.54%	35.81%
Below 10K	5,536	6,638	83.40%	22.50%
10K–15K	4,959	6,085	81.50%	20.16%
20K–25K	3,560	4,823	73.81%	14.47%
35K+	1,738	2,861	60.75%	7.06%

The **25K–35K segment contributes the most to sales volume (35.81%)**, which is primarily driven by the X300FE model. However, the **Below 10K and 10K–15K segments lead in achievement %**, both crossing 81%. These budget segments are where market demand in Northeast India is strongest, and they are meeting targets more consistently than mid or premium ranges.

The **35K+ segment is the weakest** at only 60.75% achievement — this is entirely driven by the Ultra model, which is struggling significantly across the region.

3. Branch-wise Insights

Branch	Sell Out	Target	Achievement %	Remark
Silchar	3,579	4,584	78.08%	☑ Best
Agartala	3,469	4,444	78.06%	☑
Dimapur	3,324	4,263	77.97%	
Guwahati	3,889	4,991	77.92%	Highest Volume
Imphal	3,512	4,572	76.82%	
Shillong	3,405	4,494	75.77%	⚠ Watch
Jorhat	3,424	4,568	74.96%	⚠ Lowest

Best Performing Branch: Silchar (78.08%), followed very closely by Agartala (78.06%). Silchar also has the lowest average DOS (12.70 days), meaning stock moves faster here than anywhere else in the region. This suggests better retailer engagement or stronger consumer demand in the Silchar market.

Lowest Performing Branch: Jorhat (74.96%). Despite receiving healthy inward supply (4,372 units), Jorhat's sell-out has not kept pace. This gap points to either weaker retailer coverage, lower footfall, or a need for more targeted sales activity in that area. It is worth investigating at the ground level.

Guwahati leads in absolute sell-out volume (3,889 units), which is expected given it is the largest commercial hub in the Northeast. However, its achievement percentage sits slightly below Silchar and Agartala, which may indicate that targets were set more ambitiously there.

Shillong (75.77%) is also below the regional average and worth monitoring alongside Jorhat. Overall, all branches are carrying closing stock levels that are significantly higher than their daily sell-out rates — this is worth reviewing from a working capital standpoint.

4. Model-wise Insights

Model	Sell Out	Target	Achievement %	Sales Contribution %
Y11	6,771	8,171	82.87%	27.52%
Y21	6,417	7,844	81.81%	26.08%
X300FE	4,709	6,242	75.44%	19.14%
V70FE	4,478	6,023	74.35%	18.20%
Ultra	2,227	3,636	61.25%	9.05%

Top contributor: **Y11** — 6,771 units sold, best achievement at 82.87%, and accounts for over one in every four units sold in the region. Y21 follows closely at 6,417 units. Together, Y11 and Y21 make up **53.6% of total sales** — more than half the region's volume from just two models.

X300FE shows 4,709 units sold with 75.44% achievement — a more complete and accurate picture than before. It is now the third largest contributor at 19.14%, which is a meaningful share.

Ultra is the most concerning model — only 2,227 units sold against a target of 3,636, and it carries 89,788 units in closing stock with an average DOS of over 40 days. This model is clearly not moving at the pace supply planning expected.

5. DOS and Slow-Moving Stock Insights

DOS (Days of Stock) indicates how many days the current closing stock will last at the current daily sales rate. A healthy DOS is typically 15–30 days for smartphones.

Model	Avg DOS	Max DOS	Closing Stock	Sell Out
Ultra	40.37	86.44	89,788	2,227
X300FE	13.94	30.55	65,979	4,709
V70FE	12.35	24.69	54,848	4,478
Y11	4.99	12.29	33,008	6,771
Y21	4.26	7.90	27,342	6,417

Critical: Ultra (Avg DOS 40.37 days, Max 86.44 days)

An average DOS of 40 days — with peaks reaching 86 days in some branch-level records — means that in certain locations, the current stock could last nearly three months at the current rate of sale. For a smartphone, this is a serious problem. Technology becomes outdated quickly, and holding excess stock this long ties up working capital, occupies warehouse space, and increases the risk of obsolescence or forced markdowns

X300FE and **V70FE** are within a manageable DOS range (12–14 days average), though their closing stocks are still quite high in absolute numbers. These need to be watched carefully in the next period.

Warning: Y11 (DOS 4.99) and Y21 (DOS 4.26) — Stockout Risk

With DOS of just 4–5 days on average, Y11 and Y21 are dangerously close to stockout territory. If replenishment does not happen in time, the region risks losing sales on its two best-performing models — which account for over 53% of total sell-out.

Branch-level DOS: Imphal has the highest average DOS (17.90 days) while Silchar has the lowest (12.70 days), which again reinforces Silchar as the region's most active market.

6. Stock Status Insights

Status	Records	% of Total	Key Observation
Low Stock	719	68.48%	Mainly Y11, Y21, V70FE, X300FE — fast-moving models depleting quickly
Healthy	197	18.76%	X300FE and V70FE have the most Healthy records
Overstock	134	12.76%	Almost entirely Ultra — 133 of 134 Overstock records belong to Ultra

The Low Stock label dominates at 68.48%, but this needs context — it does not mean every model is running short. The category split by model tells a very different story:

Model	Healthy	Low Stock	Overstock
Ultra	63	9	133
V70FE	55	155	0
X300FE	79	140	1
Y11	0	209	0
Y21	0	206	0

- Y11 and Y21 are almost entirely in the Low Stock category (209 and 206 records respectively, with zero Overstock entries). These models sell so quickly that stock gets depleted before it can build up. This is a genuine supply risk.
- Ultra is the opposite — 133 out of its 205 records are Overstock, with only 9 in Low Stock. This model is being supplied heavily but is not clearing off shelves.
- X300FE and V70FE show a mixed picture: mostly Low Stock or Healthy, with minimal Overstock. Their stock situation is less alarming than Ultra's, but their DOS is still in the 12–14 day range.

Overall, the stock distribution is unbalanced. The models that are selling well (Y11, Y21) are at risk of running short, while the model that is not selling (Ultra) is sitting in excess. This mismatch between supply planning and actual demand is the key operational issue to address.

7. Target Achievement Insights

Strong Performers

- Y11 — 82.87% achievement, the best among all models. Safest bet for consistent supply.
- Y21 — 81.81%, a close second. Together with Y11, holding up the overall region numbers.
- Silchar — 78.08%, highest achievement among all branches.
- Agartala — 78.06%, a very close second.
- Below 10K segment — 83.40%, the only segment crossing 83%.

Weak Performers

- Ultra model — 61.25% is significantly below the overall regional average of 77.08%.
- Jorhat branch — 74.96%, the lowest among all branches.
- Shillong branch — 75.77%, second lowest branch.
- 35K+ segment — 60.75%, matching the Ultra model's underperformance.
- 20K–25K segment — 73.81%, V70FE pulling this segment below target.

8. Inward Supply Analysis

The supply-to-sellout ratio shows clearly where supply and demand are misaligned:

Model	Total Inward	Total Sell Out	Supply : Sellout	Assessment
Ultra	6,071	2,227	2.73x	⚠ Severely oversupplied
X300FE	7,030	4,709	1.49x	Moderate — watch closing stock
V70FE	6,073	4,478	1.36x	Moderate — acceptable range
Y11	6,190	6,771	0.91x	☑ Undersupplied — stockout risk
Y21	5,976	6,417	0.93x	☑ Undersupplied — stockout risk

Ultra is being supplied at **2.73 times its sell-out rate**. For every unit sold, nearly three are arriving. This is the clearest evidence of a supply planning issue — and the root cause of the 89,788-unit closing stock pile-up.

Y11 and Y21 have ratios below 1.0, meaning they are selling more than they are receiving. They are drawing down on opening stock to fulfill demand. If this continues without correction, stockouts become likely.

At the branch level, every branch received more inward supply than it sold — which is normal inventory practice — but the resulting closing stocks are elevated across all branches, suggesting that demand forecasting may need to be tightened for the next planning cycle.

9. Data Quality Observation

Extra Column Added: Remark (Not in Original Task Specification)

During the data cleaning process, a Remark column was added as a proactive data validation step. This was not part of the original task requirements, but it was included to improve the reliability of all downstream MIS calculations.

Out of 1,050 records, 9 were flagged as "Need Verification" because they showed negative closing stock values. All 9 records belong to the Y11 model, spread across Guwahati and Silchar branches, occurring between 16th and 30th May 2026.

Why Negative Closing Stock is a Problem

Logically, closing stock cannot be negative. The formula is:

$$\text{Closing Stock} = \text{Opening Stock} + \text{Inward Supply} - \text{Sell Out}$$

A negative result means sell-out recorded is greater than total available stock — which is not physically possible in standard inventory tracking. Possible causes include:

- Data entry errors in sell-out or opening stock figures
- Missing inward supply entries — stock that physically arrived but was not recorded in the system
- Stock mismatch between system records and physical inventory count
- Incorrect sell-out recording, such as double-counting or entering return quantities incorrectly

10. Final Business Recommendations

1. Pause or significantly reduce Ultra inward supply. With 89,788 units in closing stock and a DOS of 40 days, this is the most urgent inventory issue in the dataset. No new supply should come in until existing stock reaches a healthy level.
2. Prioritise Y11 and Y21 replenishment immediately. With a DOS of just 4–5 days, any delay in supply could cause stockouts for the region's top two models, which account for 53.6% of all sales.
3. Conduct a branch-level review for Jorhat and Shillong. Both are underperforming despite receiving adequate supply. A review of retailer engagement, sell-out schemes, and local market conditions is recommended.
4. Re-evaluate the Ultra go-to-market strategy for Northeast India. At 61.25% achievement over 30 days, the problem may go beyond supply. Pricing, marketing visibility, and consumer awareness for the premium segment in this geography may need to be revisited.
5. Resolve the 9 negative closing stock records. These should be reviewed with branch operations teams and corrected in the system. Until resolved, Y11 figures for Guwahati and Silchar during the second half of May remain unreliable.
6. Tighten supply planning for V70FE and X300FE. Both have acceptable DOS currently, but their closing stocks are large. If demand softens, they may slide toward an overstock situation similar to Ultra.
7. Use segment-level achievement data for target setting. The data shows a clear difference between what the Below 10K / 10K–15K segments can achieve (81–83%) versus what the 35K+ segment is achieving (60.75%). Future targets should reflect these market realities.